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Louis Ledoux

Associate Professor – Maître de conférences

"Post Hoc Ergo Propter Hoc"

Research Interests

- **Hardware Design & Synthesis:** Floating-Point Units, Systolic Arrays, Tensor Accelerators, FPGAs, ASIC Design, RTL generation
- **Computer Arithmetic:** Number Representations, FloPoCo, Posit & IEEE 754 Standards, Fixed-Point Polynomial Approximations, Accuracy–Energy Tradeoffs
- **High Performance Computing:** BLAS, GEMMs, Heterogeneous Workloads, Numerical Analysis, Supercomputing
- **End-to-End EDA Flows:** From mathematical problem to silicon implementation, integrating MLIR-based compilation, RTL generation, and open-source silicon flows

Experience

2026 – Now **Associate Professor (*Maître de conférences*)**, *Université de Rennes – ISTIC*, Rennes

Teaching-and-research faculty member (*enseignant-chercheur*) at ISTIC, with research in IRISA's Architecture Department (D3) as a member of the Inria Taran project-team.

- Contributing to the design and tapeout of AI accelerators within the national PEPR CAMELIA program.
- Designing systolic-array and tensor architectures with small-format floating-point arithmetic.

2025 – 2026 **Postdoctoral Researcher**, *INSA/INRIA – Emeraude*, Lyon

Worked on arithmetic compilation and optimization within the *Emeraude* team, at the intersection of *MLIR*, *HAriCo/FloPoCo*, *CIRCT*, *Faust*, and open EDA/hardware flows. The research focused on bridging high-level mathematical reasoning with low-level hardware synthesis and reproducible compiler-to-silicon validation.

- Developed multi-level arithmetic transformations and MLIR dialects for fixed-point approximation, ASIC, and FPGA targets.
- Automated hardware generation from DSP code and contributed to FloPoCo's refactoring toward explicit IR-based circuit generation.
- Presented results at *DSD 2025*, *EuroLLVM 2025*, the *PEPR IA Embarquée* workshop in Aussois (2026), and *EuroLLVM 2026*.

2018 – 2024 **Researcher**, *Barcelona Supercomputing Center (BSC) - RoMoL/CAOS/SONAR*, Barcelona

As a BSC researcher during my PhD, I published peer-reviewed papers, attended international conferences, and developed software toolchains and hardware designs for numerical acceleration. The conducted research is summarized as follows:

- Explored co-designed hardware/software acceleration of posit arithmetic for FPGA and ASIC technologies, and POWER9 hosts.
- Developed Kulisch/Quire accumulators for any floating-point representation.
- Conducted accuracy and energy budgeting tailored to workload numerical requirements.
- Designed Systolic Array architecture for HPC workloads with three directions of data flow, infinite number representations, and tailored internal precision.
- Proposed serialized, area-efficient division units targeting the SIMD/Vector paradigm at the architecture level, further improving the parallelism/latency/energy ratio.

2017 - 2018 **Hardware Engineer**, *b⟨⟩com*, Rennes

Coop student under a professionalization contract during my final year of study. Engaged in one year of R&D focused on FPGA acceleration in the cloud, aiming to evaluate the feasibility of these novel solutions.

- Successfully integrated an IP (real-time SDR to HDR) to convert video from NVMe to NVMe through cloud FPGA.
- Developed the IP integration using HDLs and tweaked PCIe drivers (EDMA and XDMA).
- Scheduled user processes asynchronously with OpenCL to overlap reads and writes, successfully saturating PCIe bandwidth (≈15.8 GB/s).
- Traveled to Xilinx XDF Frankfurt 2018 to conduct operational monitoring and analysis.

Keywords: SDAccel, OpenCL, Xilinx FPGA (UltraScale VU9P), VHDL, SystemVerilog, AWS F1 instances, Linux driver, Linux kernel, PCIe, C++1x, C, DMA, AaaS (Acceleration as a Service), FaaS (FPGA as a Service), Virtual Machine, Docker, PCIe virtual functions.

July 2017 **Backend Developer**, *WaryMe*, Rennes

Summer internship focused on developing the entire back end of a people security application.

- Designed and implemented backend services and APIs.
- Ensured secure data transmission using HTTPS and Let's Encrypt.
- Deployed and managed the application on AWS with PM2 and Nginx.
- Automated deployment processes with Jenkins.
- Collaborated with front-end developers using Angular and TypeScript.

Keywords: Node.js, Git, C++, C, Angular, TypeScript, HTTPS, Let's Encrypt, PM2, Nginx, AWS, Jenkins.

July 2016 **Backend Developer**, *ASKIA*, Paris 10ème

During this summer internship, I developed an automated CLI tool for publishing surveys on popular platforms such as GitHub and Zendesk. **Key Concepts:** Node.js, HTTP(S), REST API, Git, Test-Driven Development (TDD), Event-Driven/Asynchronous Programming, Jasmine Framework, Mocks, Stubs.

July 2014 **Electronics Technician**, *Radio Electronique Rennaise (R.E.R)*, Rennes

As an electronics technician, I was responsible for repairing various electronic devices from customers, with an emphasis on audio equipment. Tasks included soldering, reverse engineering amplifier circuits, welcoming customers, and sorting/ordering electronic parts.

Education

2018 – 2024 **PhD Student**, *Universitat Politècnica de Catalunya (UPC)*, Barcelona

Supervisor: Marc Casas Guix. Thesis title: "Floating-Point Arithmetic Paradigms for High-Performance Computing: Software Algorithms and Hardware Designs". Jury members attributed the **excellent** grade. • HAL:tel-04754167.

2015 – 2018 **Engineering School**, *ESIR*, Université de Rennes

Completed a three-year program culminating in an **Engineer diploma** certified by the CTI (Comité des Titres d'Ingénieurs) and a **Master's degree** (Magister) in Computer Science.

2013 – 2015 **Classe Préparatoire**, *CUPGE ESIR*, Université de Rennes

Intensive program with a strong emphasis on Mathematics and Computer Sciences.

Teaching *(University teaching load: **100h**)*

13 Aug. 2026 **Course Instructor – Digital Design with MLIR and CIRCT**, *ACM Europe School on MLIR 2026*, A Coruña, Spain

Designed and delivered a 1h45 interactive course for an international audience, using quizzes and guided exercises to teach digital-design fundamentals and practical MLIR/CIRCT flows: IR lowering, optimization, equivalence checking, logic synthesis, and SystemVerilog generation. Developed a capstone in which participants lowered E4M3 floating-point multiplication from `arith.mulf` to `hw/comb` circuitry; demonstrated the end-to-end path from MLIR input to a small Sky130 ASIC GDSII layout, alongside a taped-out TinyTapeout GF0P2 design. Programme • Course slides • source and exercises.

2026 **Computer Architecture (ARC)**, *INSA Lyon*, Lyon

Practical sessions in computer architecture for the Telecommunications curriculum, covering instruction execution, pipelining, cache memories, performance evaluation, and RISC-V assembly programming.
Level: 1st year of engineering school (Telecommunications track, 3TC); Teaching load: 24h

2026 **Operating Systems (SYS)**, *INSA Lyon*, Lyon

Core operating system mechanisms, including kernel and system calls, process scheduling, virtual memory, concurrency, and file systems.
Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 16h

2026 **Long-Term Project in Compilation (PLD-COMP)**, *INSA Lyon*, Lyon

Supervision of long-term group projects on compiler design and implementation, covering intermediate representations, program analysis, and backend code generation.
Level: 2nd year of engineering school / Master 1 (Computer Science, INSA 4IF); Teaching load: 36h

2025 – 2026 **Computer Architecture (AO)**, *INSA Lyon*, Lyon

Problem-solving and practical sessions on processor organization and instruction set architectures, based on the MSP430 microcontroller, including instruction execution, addressing modes, and memory hierarchy.
Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 12h

2025 – 2026 **Digital Design (AC)**, *INSA Lyon*, Lyon

Exercises and guided sessions on digital circuit design, including combinational and sequential logic, finite state machines, and introductory hardware description languages.
Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 12h

International Peer-reviewed Journals

[Bar+26a] B. Barbe, **L. Ledoux**, A. Volkova, and F. de Dinechin, "Reconfigurable constant multipliers: hardware models, optimization algorithm and applications," *Microprocessors and Microsystems*, vol. 122, article 105270, June 2026. • DOI • HAL:05510936.

[Pop+25] M. Popoff, R. Michon, T. Risset, P. Cochard, **L. Ledoux**, S. Letz, Y. Orlarey, and F. de Dinechin, "Frugalité et conception de circuits pour le traitement du signal audio numérique," *Revue Francophone d'Informatique et Musique*, special issue "Frugalité, pérennité et création," vol. 11, 2025. • DOI • HAL:05489376.

[Led+25b] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Towards optimized arithmetic circuits with MLIR," *WiPiEC Journal* (Works in Progress in Embedded Computing), vol. 11, no. 1, 2025, Salerno, Italy. Associated conference presentation at the Euromicro Conference on Digital System Design (DSD). • DOI • HAL:05385229.

International Peer-reviewed Conference Papers

[Bar+26b] B. Barbe, **L. Ledoux**, F. de Dinechin, and A. Volkova, "RSCM-Net: FPGA-Aware Co-Design of Quantized DNNs with Reconfigurable Single Constant Multipliers," *International Conference on Field Programmable Technology (FPT 2026)*, Honolulu, Hawaii, USA, December 7–10, 2026. *Forthcoming.* • Conference.

[Led25] **L. Ledoux**, "Design-space exploration of serialized floating-point division for DLP architectures," in *28th Euromicro Conference Series on Digital System Design (DSD 2025)*, Salerno, Italy. • DOI • HAL:05385247.

[LC23a] **L. Ledoux** and M. Casas, "An open-source framework for efficient numerically-tailored computations," in *33rd International Conference on Field-Programmable Logic and Applications (FPL 2023)*, Gothenburg, Sweden. • DOI • arXiv:2406.02579 • HAL:04277512.

[LC22] **L. Ledoux** and M. Casas, "A generator of numerically-tailored and high-throughput accelerators for batched GEMMs," in *30th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2022)*, May 2022, pp. 1–10, New York, USA. • DOI • HAL:04103774.

Tutorials, Workshops & Demos

MLIRW26 **7th MLIR Workshop @ EuroLLVM 2026**, Dublin, Ireland, April 13, 2026. *Talk.* Progressive Arithmetic Lowering from Tensor Kernels to Synthesizable Datapaths. • HAL:05594483.

ARITH26 *Tutorial: Optimizing Datapath Circuits with MLIR and CIRCT*, with Sam Coward, June 27, 2026. Exercises and slides.

Demo: Printing Mathematical Intent onto Silicon with Emeraude-MLIR, HAriCo, and OpenROAD, June 29, 2026, Fulda, Germany. Programme.

Contributions to Publications

[Ven24] Contributor (not a listed author) to: M. Venn, "Tiny Tapeout: a shared silicon tapeout platform accessible to everyone," *IEEE Solid-State Circuits Magazine*, 2024. Initiated manuscript drafting, ensured academic formatting, and contributed to revision. • IEEE article • manuscript commits.

Poster Presentations

[Led26e] **L. Ledoux**, "HAriCo: Floating-Point MLIR to CIRCT Datapaths," in *ACM Europe School on MLIR 2026*, A Coruña, Spain, August 10–14, 2026. • HAL:05725118.

[Led26d] **L. Ledoux**, "Multi-Level Compilation of AI Tensors and Activation Functions to Silicon," in *PEPR IA Days*, Rennes, France, June 22–24, 2026. • HAL:05671676.

[Led+26c] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Progressive Arithmetic Optimizations: Tensor and DSP Kernels to Synthesizable Datapaths," • *3rd FPGA Developers' Forum (FDF) meeting*, CERN, May 27–29, 2026. • HAL:05631193v1.

[Led+26b] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Floating-Point Datapaths in CIRCT via HAriCo and harico-arith-to-comb Lowering," in *EuroLLVM Developers' Meeting 2026*, Dublin, Ireland. • HAL:05576131.

[Led+26a] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Arithmetic lowering with Emeraude-MLIR: bridging tensor and DSP kernels to silicon datapaths," Aussois, France. • HAL:05489427.

[Led+25a] P. Cochard, L. Forget, F. de Dinechin, and **L. Ledoux**, "Towards multi-level arithmetic optimizations," in *EuroLLVM 2025*, Berlin, Germany. • HAL:05063466.

[LC24b] **L. Ledoux** and M. Casas, "LLMMMM: large language models matrix–matrix multiplications characterization on open silicon," in *11th BSC Severo Ochoa Doctoral Symposium 2024*, Barcelona, Spain. • HAL:04592229.

[LC24a] **L. Ledoux** and M. Casas, "The grafted superset approach: bridging Python to silicon with asynchronous compilation and beyond," in *4th Workshop on Open-Source Design Automation (OSDA 2024)*, hosted at *DATE*, Valencia, Spain. • HAL:04587458.

[LC23b] **L. Ledoux** and M. Casas, "Open-source GEMM hardware kernels generator: toward numerically-tailored computations," in *10th BSC Severo Ochoa Doctoral Symposium 2023*, Barcelona, Spain. • arXiv:2305.18328 • HAL:04094835.

Invited Talks & Seminars

[LC25] P. Cochard and **L. Ledoux**, "FloPoCo and MLIR: a multi-level compilation framework for many intents," Holigrail Seminar, Sorbonne University, Paris, France, September 18, 2025. • HAL:05412130.

[Led24] **L. Ledoux**, "The walls and the dark silicon era: an arithmetic perspective," seminar at Inria Rennes (Team TARAN), Rennes, France, December 2, 2024. • HAL:04823130.

[LC19] **L. Ledoux** and M. Casas, "Accelerating DL inference with (Open)CAPI and posit numbers," in *OpenPOWER Summit Europe 2019*, Lyon, France. • HAL:04094850.

Educational Videos & Scientific Outreach

Zero to ASIC **Arbitrary-precision floating-point systolic arrays.** Educational presentation of a systolic-array generator with configurable floating-point formats and accumulator precision, for mentoring and teaching in the Zero to ASIC and Tiny Tapeout communities. • Educational video (YouTube).

Zero to ASIC **Visualization-oriented global placement.** Educational presentation for Matt Venn's Zero to ASIC course on shaping a chip during global placement while exposing the underlying physical-design process. • Lecture recording (Google Drive).

SUF demo **64×64 systolic-array placement.** Video demonstration of the SUF flow's placement visualization for a 4,096-PE array in ASAP7. • YouTube demonstration.

Moduland **Moduland festival (GRAME–cncm).** Staffed a booth for nine hours on behalf of the Emeraude team (INSA–Inria–GRAME), presenting an electronic instrument and discussing links between sound, numerical computing, and integrated-circuit design. • Instagram.

Conference Peer Reviewing

2026 FCCM 2026 (×1), ASAP 2026 (×2), and ICECS 2026 (×1).

Conference Organization Experience

LAC25 **Linux Audio Conference**, Lyon, France (2025). *Technician.* Responsible for sound, YouTube streaming, and microphone management. event website.

JIM25 **Journées de l'Informatique Musicale**, Lyon, France (2025). *Technician.* Responsible for sound, YouTube streaming, and microphone management. event website.

HC32 **HotChips 32nd**, Remote (Digital), SARS-CoV-2 Period (2020). *Remote Assistant.* Assisted with digital entrance management, conference link distribution, and attendee support during the event. event website.

FPL29 **Field-Programmable Logic Conference 29th**, Barcelona, Spain (2019). *Reception and Entrance Coordinator.* Managed reception desk activities, including badge distribution and providing microphones for speakers during presentations. event website.

Research Project & Internship Mentoring

2026 – 2027 **Thomas Brembilla** (ENS Rennes, M2) and **Edouard Pierre** (ISTIC, M1 Electronics). *Ongoing co-supervision* with Noë Mercier-Brosse and Olivier Sentieys for the 7th Thales RISC-V student competition: acceleration of the QOA audio encoder on the CVA6 RISC-V processor. • 2026–2027 announcement.

2026 – 2027 **ENS Rennes – M1 SIF research project (ongoing)**. "Vector Floating-Point Architectures for Softmax and Exponential Acceleration," co-supervised with Sami Ben Ali and Olivier Sentieys in the TARAN team (Inria/IRISA). Study of vector FPUs and exponential approximations, balancing softmax throughput, hardware cost, and numerical accuracy within CAMELIA. • Project topic (PDF) • ENS project listing.

2026 **Esteban Wybouw**, "Porting FloPoCo to HAriCo: an IR-based core generation framework for multi-HDL hardware generation," co-supervised with Florent de Dinechin and Pierre Cochard. Internship completed in August 2026.

2026 **Azure Blondon, Hugo Berthet-Rambaud, Reda Id Hammouche, and Simon Cabias**. L3 / first year of the engineering cycle (INSA Lyon, 3IF): research internships in the FloPoCo/HAriCo framework. Personal supervision share: approximately 15%, alongside lead supervisors Pierre Cochard and Florent de Dinechin.

Academic Summer Schools and Workshops

2026 **ACM Europe School on MLIR**, A Coruña, Spain, August 10–14. *Instructor.* Delivered "Digital Design with MLIR and CIRCT" (1h45); see Teaching for course materials. Programme.

2021 **ACM Europe Summer School on High Performance Computing**, Barcelona, Spain. *Participant.* Attended lectures and hands-on sessions covering exascale architectures, programming models, and numerical methods. school website.

2019 **Yale Patt's Computer Architecture Summer School**, Barcelona, Spain. *Participant.* Attended the special summer edition of Prof. Yale Patt's course on out-of-order execution and advanced microarchitectural design.

2019 **Xilinx Machine Learning and Artificial Intelligence Seminar**, Paris, France. *Participant.* Technical seminar focused on machine learning workloads and AI acceleration platforms.

2019 **PRACE Summer School on Heterogeneous Programming with OmpSs@FPGA**, Barcelona, Spain. *Participant.* Training on FPGA-based acceleration and heterogeneous programming using the OmpSs@FPGA model.

Technology Development: Software & Hardware

Software SUF: OpenROAD-based ASIC design-space exploration and flow orchestration; OSFNTC: numerically-tailored GEMM generator with PyTorch/NumPy/OpenBLAS/OpenCAPI integration; MLModelMLIRGEN: PyTorch/`torchvision`→MLIR exporter.

Hardware IP POF: SystemVerilog Posit operators, datapaths, quires, and verification benches; FloPoCo: Virtex UltraScale+ timing model, systolic-array generator work, and HAriCo/IR-oriented refactoring.

Utilities VH2V: VHDL→Verilog bridge for open ASIC flows; OpenROAD GPL: placement visualization and chip-shaping experiments; gdsiistl: GDSII→STL layout visualization; Codez: Typst/CeTZ annotated code figures with MLIR/Python/SystemVerilog support.

Upstream CIRCT PR9263 and PR9245: arithmetic lowering and HDL-oriented transforms; oc-accel and capi2-bsp: OpenCAPI/CAPI2 accelerator support; OpenROAD PR: Arch/Manjaro installer support.

Tapeouts MPW5 Sky130: posit/Quire systolic array; TT GF0P2: Faust→MLIR→silicon soft-clipper; TT SKY26A: MLIR→floating-point accumulator; IHP BF16 and IHP IEEE754: CMOS5L floating-point lowering demonstrators; TT SKY25B: modified-placement demonstrator; MPW1 and MPW4: integration contributions.

Languages

French Native

Spanish Native (with an honest French accent)

English Full Proficiency